

Supplemental Material for “Learning impurity incorporation energetics across host and impurity chemistry in two-dimensional materials with a defect-aware graph Transformer”

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I. DATA AUDIT AND FIXED EVALUATION PARTITIONS

The retained modeling set is obtained through a sequential source audit, target-provenance check, unique-impurity requirement, and duplicate-control procedure. Figure S1 summarizes those exclusions, the formation-energy distribution, chemical coverage, and the fixed evaluation partitions used throughout the study.

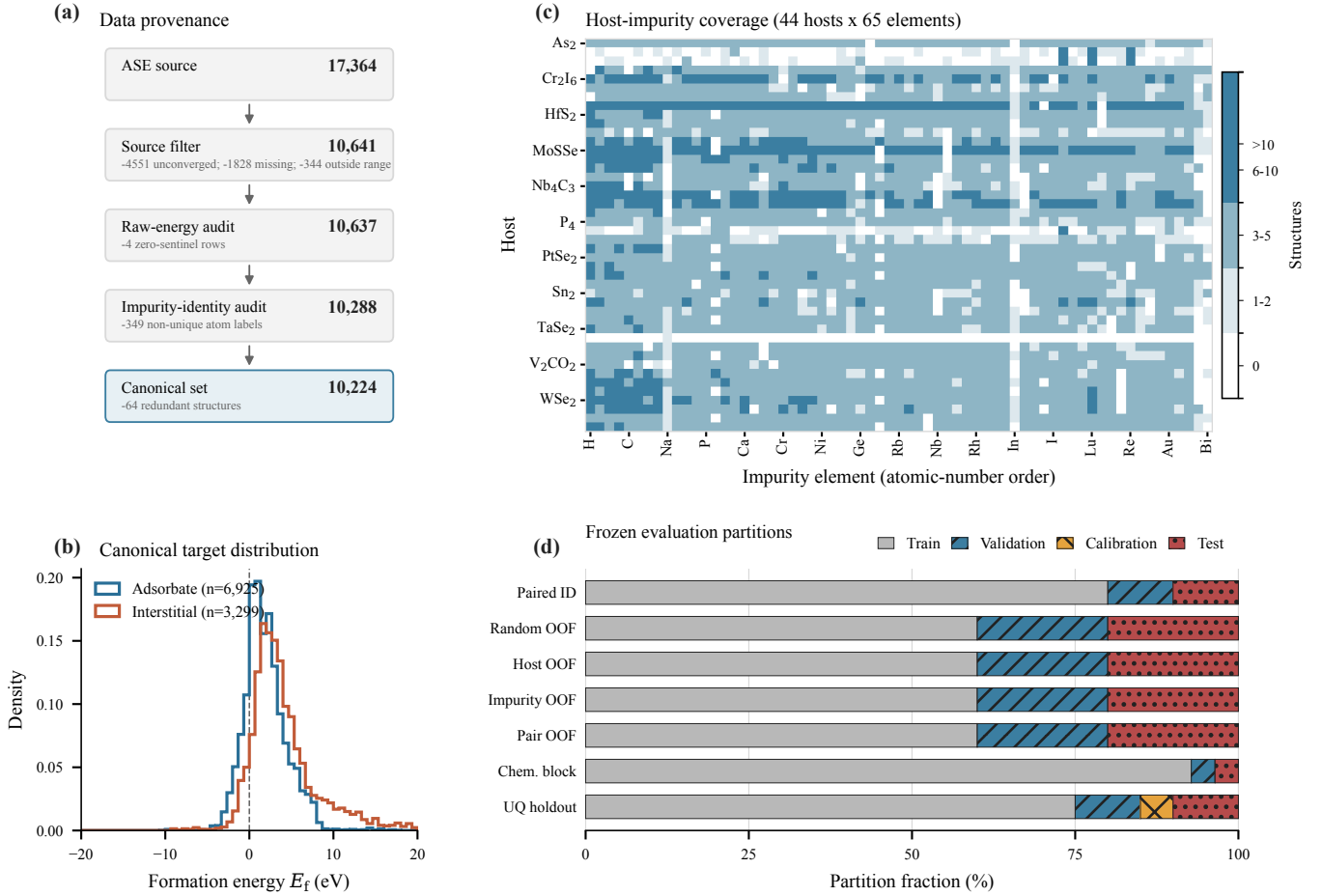


FIG. S1. Data provenance and evaluation protocol. (a) Sequential source filtering, raw-energy reconstruction audit, impurity-identity audit, and structure canonicalization. The final set excludes four rows with zero sentinels in a raw energy component, 349 rows without a unique impurity atom, and 64 redundant structures. (b) Formation-energy distributions for the two incorporation classes in the retained set. (c) Number of structures for each of 44 hosts and 65 impurity elements; white cells are unobserved combinations. (d) Train, validation, calibration, and test fractions for each evaluation regime. Bars for fivefold protocols show fold averages. All regimes cover the same 10,224 retained structures; the common 417-row exclusion set is omitted from the bars.

II. ARCHITECTURE FACTORIAL VISUALIZATION

The paired architecture experiment is tabulated in the main text. The corresponding visual comparison is retained here so that the main-text figure sequence can emphasize chemical transfer and materials-physics analysis.

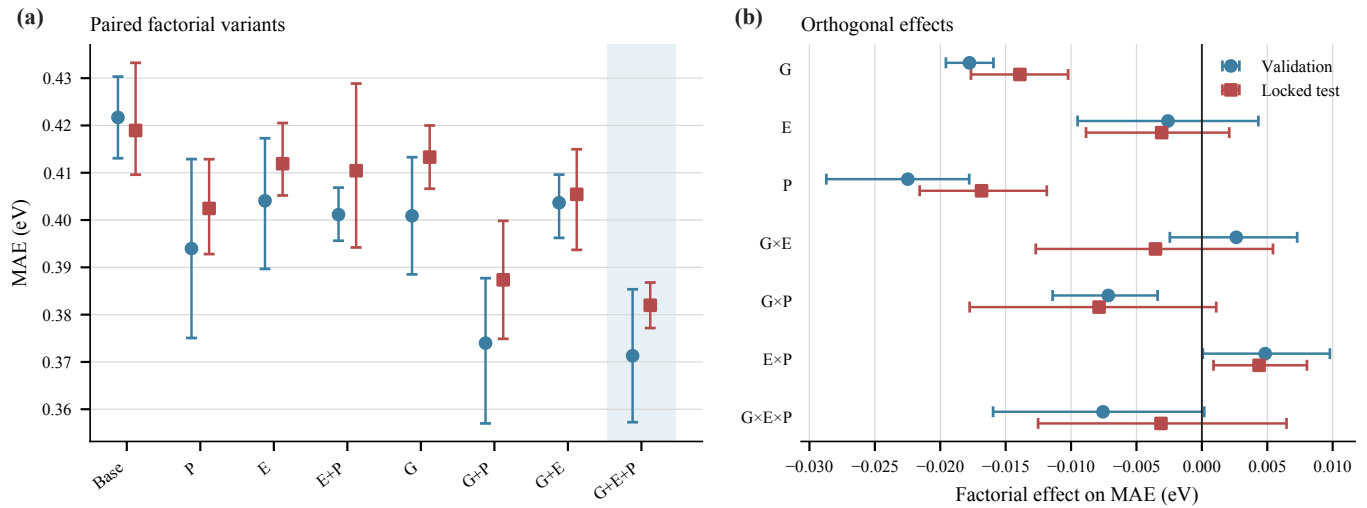


FIG. S2. Paired architecture experiment. (a) Validation and test MAE for all eight G/E/P variants; shading marks the validation-selected variant. (b) Orthogonal main effects and interactions on validation and test MAE, with 95% bootstrap intervals over paired repeats. Negative effects reduce MAE.

III. REPRODUCIBILITY SPECIFICATION

Table SI records the fixed model and optimization settings used in the main comparison. The resolved configuration for every run, including split and seed, is archived under `configs/prm/`; the corresponding run manifests bind those configurations to checkpoints, predictions, environments, and output hashes.

TABLE SI. Fixed settings for the selected DART model and the predefined periodic SchNet comparator. Settings not shown as different were shared.

Setting	DART (g111)	Periodic SchNet (additive readout)
Atomic input	Fixed 128-dimensional ct-UAE vector, nine tabulated elemental attributes, and a learned 128-channel impurity-status embedding	Learned atomic-number embedding
Interaction stack	Three 5-Å local layers with 32 radial and 32 angular functions, followed by two four-head geometric Transformer blocks with 32 radial-bias centers over 0–12 Å and all-atom attention	Six 5-Å interaction blocks with 50 Gaussian functions
Hidden width and readout	128 channels; gated fusion of an atom-attention sum and channelwise maximum	128 channels and filters; atomwise additive scalar readout
Trainable parameters	820,558	455,937
Optimization	AdamW, batch size 64	AdamW, batch size 32
Shared schedule	150 epochs; MAE objective; learning rate 5×10^{-5} to 5×10^{-4} over a ten-epoch linear warmup, then cosine decay to 10^{-6} ; weight decay 10^{-4} ; gradient clipping at 5.	
Shared sampling and regularization	Training-target standardization; host-balanced sampling of 200 structures per host per epoch; 0.03-eV Gaussian target noise; no coordinate or strain augmentation. DART additionally uses 0.1 dropout.	

Factorial assignment seeds were 42–46 with paired model seeds 142–146. Transfer runs used DART seeds 242–244 and SchNet seeds 342–344, with one fixed seed per fold for random and host–impurity-pair cross-validation and three seeds per split for host, impurity, and chemistry-block evaluation. The dedicated DART uncertainty ensemble used seeds 442–446.

The descriptor search was also fixed before test aggregation. Ridge used $\alpha \in \{0.01, 0.1, 1, 10, 100\}$. Random forests used 400 trees with `max_features` in $\{0.5, 1.0\}$ and `min_samples_leaf` in $\{1, 2\}$. Histogram gradient boosting used 400 iterations, learning rate 0.05, leaf counts in $\{31, 63\}$, and L_2 regularization in $\{0, 1\}$. LightGBM used 600 trees, learning rate 0.05, leaf counts in $\{31, 63\}$, child counts in $\{20, 50\}$, L_2 regularization 1, and row/column subsampling fractions 0.9. A hyperparameter setting and then a learned family were selected independently inside each split by validation MAE; the training-target mean predictor was reported separately and never entered family selection.

IV. INTERNAL UNCERTAINTY AND SELECTIVE PREDICTION

The internal ensemble analysis uses a dedicated calibration/test partition drawn from the retained IMP2D population. Its intervals therefore assess marginal calibration inside that population rather than calibration under an explicit host or impurity shift.

TABLE SII. Internal split-conformal interval performance on the dedicated test partition. These structures were excluded from ensemble fitting, checkpoint selection, and calibration, but not from the earlier architecture-development corpus. Coverage is in percent and mean width is in eV.

Nominal	Observed	Mean width	Quantile
50	52.0	0.336	0.270
80	78.3	0.752	0.604
90	87.2	1.141	0.917
95	93.7	1.847	1.484

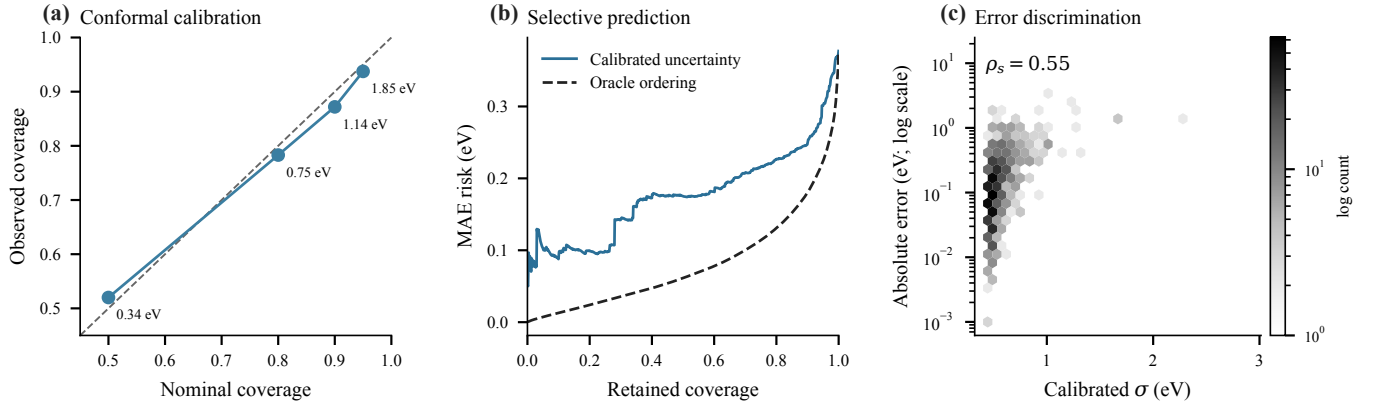


FIG. S3. Internal uncertainty evaluation on the dedicated ensemble test split. (a) Nominal versus observed split-conformal coverage; annotations give mean interval widths. (b) MAE risk as increasingly uncertain predictions are rejected, compared with oracle ordering by realized error. (c) Calibrated standard deviation versus absolute error on the dedicated test partition.

V. RETROSPECTIVE RERANKING METRICS

These metrics compare predictions and reference energies over available DFT-relaxed IMP2D geometries. The geometries and reference energies were produced by the same completed DFT workflow, so the analysis does not replace candidate generation or relaxation.

TABLE SIII. Retrospective out-of-fold reranking metrics from host–impurity-pair folds. Intervals are 95% nonparametric cluster-bootstrap intervals over host–impurity pairs. Accuracy gains are model minus reference in percentage points.

Metric	Model [95% interval]	Reference	Gain [95% interval]	n
Incorporation-class preference accuracy (%)	91.0 [89.7, 92.4]	66.2	24.8 [22.4, 27.2]	1730
Global exact-site accuracy (%)	69.5 [67.3, 71.7]	23.4	46.1 [44.0, 48.3]	1730
Global reranking regret (eV)	0.139 [0.113, 0.169]	–	–	1730
Within-class exact-site accuracy (%)	73.6 [72.0, 75.1]	37.1	36.5 [35.0, 38.0]	3140
Within-class top-2 accuracy (%)	91.4 [90.2, 92.6]	62.1	29.2 [28.0, 30.4]	2145
Within-class reranking regret (eV)	0.103 [0.087, 0.121]	–	–	3140

VI. DESCRIPTIVE ERROR HETEROGENEITY

Across host and impurity groups, the Spearman associations between group sample count and group MAE were -0.523 and -0.001 , respectively. The largest group MAEs occurred for host Bi₂I₆ (1.851 eV; $n = 52$) and impurity In (1.502 eV; $n = 35$). Because group size covaries with chemistry, these are descriptive associations rather than evidence that sample count alone causes the observed errors.

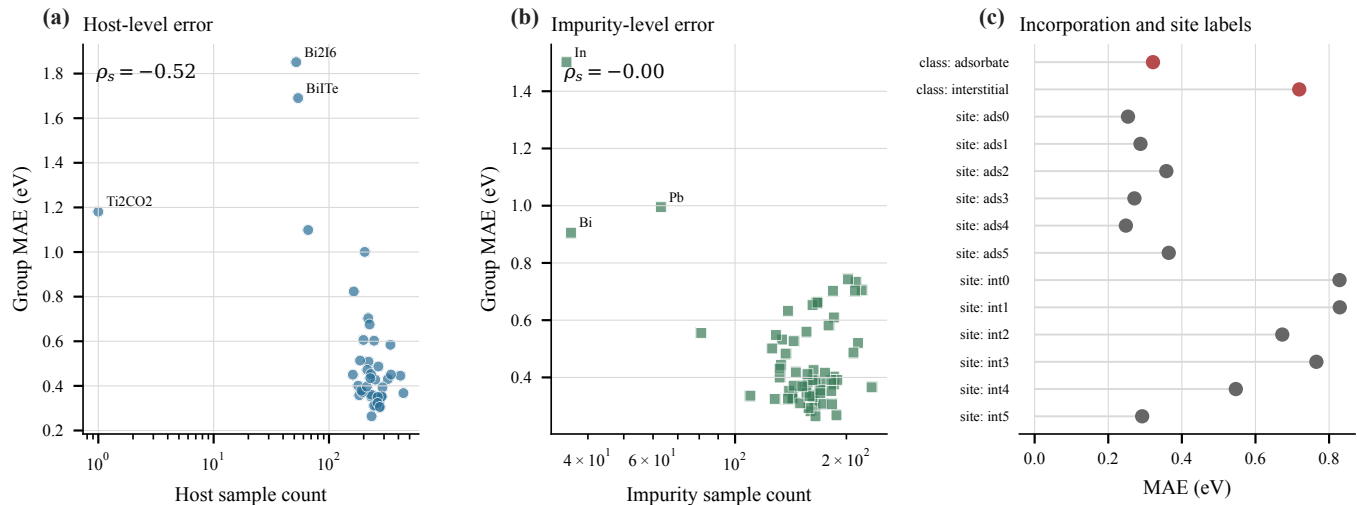


FIG. S4. Error heterogeneity from DART pair-held-out out-of-fold predictions. (a,b) Host- and impurity-group MAE versus the number of retained structures; labels mark the three largest group MAEs and ρ_s is the descriptive Spearman association. (c) MAE by incorporation class and source site label. These summaries were not used for architecture, hyperparameter, or model selection.

VII. POST-HOC SCHNET GRAPH-READOUT SENSITIVITY

The predefined periodic SchNet comparator follows the original energy-oriented construction [1]: its learned atom-wise scalar contributions are summed to produce the graph prediction. The present defect formation energy is non-extensive with respect to host atom count, and prior atomistic studies show that graph pooling can materially affect intensive or localized quantum-property prediction [2]. Because the target may still contain long-range host response, this comparison tests the specific SchNet readout rather than assuming strict spatial locality. We therefore performed one bounded post-hoc sensitivity analysis after completing the main comparison.

The intervention changed only the graph readout from atomwise addition to mean pooling. It retained all five predefined host-held-out folds, seeds 342–344, the SchNet interaction stack, learned atomic-number input, optimizer, learning rate schedule, host-balanced sampler, target normalization, target-noise stream, gradient clipping, and 150-epoch budget. Predictions were averaged over the three seeds within each fold. The pooled mean-minus-add difference in absolute error was evaluated with 20,000 bootstrap samples clustered by the 44 host materials.

TABLE SIV. Post-hoc SchNet graph-readout sensitivity under the fixed host-held-out protocol. Each fold prediction averages seeds 342–344. The difference is mean-readout MAE minus additive-readout MAE, in eV. The pooled host-cluster bootstrap 95% interval for this difference is [-7.704, -1.256] eV. The prespecified additive-readout result remains the main comparator.

Partition	n	Add MAE	Mean MAE	Mean – add
Fold 1	2023	3.882	2.179	-1.704
Fold 2	2019	8.199	2.068	-6.132
Fold 3	2069	5.484	2.530	-2.954
Fold 4	2057	5.466	2.468	-2.998
Fold 5	2056	10.474	3.501	-6.973
Pooled OOF	10224	6.703	2.552	-4.151

Mean pooling reduced the host-held-out MAE in all 5/5 folds. Across individual samples, the Spearman association between atom count and absolute error changed from 0.278 for additive pooling to 0.010 for mean pooling. Across the 44 hosts, the corresponding association between median atom count and host MAE changed from 0.312 to 0.218. These descriptive associations are consistent with a size-sensitive additive readout, but atom count covaries with host chemistry and therefore does not identify a standalone causal mechanism. The intervention establishes that readout choice contributes to the extreme additive-SchNet error; it neither replaces the predefined main comparator nor attributes the remaining DART-SchNet difference to a single component.

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- [1] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller, SchNet – a deep learning architecture for molecules and materials, *The Journal of Chemical Physics* **148**, 241722 (2018).
 [2] D. Buterez, J. P. Janet, S. J. Kiddle, D. Oglic, and P. Liò, Modelling local and general quantum mechanical properties with attention-based pooling, *Communications Chemistry* **6**, 262 (2023).